



XAI REPORT

**Topological Uncertainty :
Monitoring trained neural networks
through persistence of activation graphs**

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1 Presentation of the article

1.1 Problem statement

The starting point of the paper [2] is a practical issue that appears whenever a trained neural network is deployed in a real decision pipeline. Even if a model reaches high training and test accuracy, this does not guarantee that its predictions remain reliable once it is used in an open-world setting. In practice, the data encountered at deployment time may differ from the training distribution, may be corrupted, or may simply be far from the examples seen during training.

In such situations it is not enough for the network to output a class label: we also need a way to assess whether its prediction should be trusted. This question is particularly important in high-stakes classification settings, where a wrong but highly confident decision can have significant consequences. The central objective of the paper is therefore to design a post-training criterion that can indicate whether a new input is processed by the network in a reliable or in an anomalous way.

The key point emphasized by the authors is that modern neural networks are often overconfident. In particular, they may assign a very high confidence score to samples that are far from the training distribution. As a consequence, monitoring only the predicted label is clearly insufficient, and even standard confidence estimates may fail. The paper therefore seeks a richer notion of uncertainty, based not only on the output layer, but on the way the entire network is activated by an input.

1.2 State of the art and baseline

The baseline considered in the paper is the standard confidence-based approach introduced in [1]. In this setting, the confidence of a prediction is defined as the maximum softmax probability:

$$\text{Conf}(x) = \max_{1 \leq k \leq K} F(x)_k,$$

where $F(x) \in \mathbb{R}^K$ denotes the network output after the softmax layer.

The underlying intuition is simple: if an input x is in-distribution, then the network should usually assign a larger confidence to its prediction than it would for an out-of-distribution input. Hence, low confidence is used as a proxy for anomaly detection. This baseline is attractive because it is extremely easy to implement and, most importantly, it works in a purely post-training setting: one does not need to retrain the model, modify the architecture, use auxiliary data, or introduce additional calibration procedures.

However, this approach has a well-known limitation. Softmax confidence is not a properly calibrated measure of uncertainty, and neural networks can remain highly confident even when they are wrong or when the input is clearly atypical. In other words, the baseline only uses information from the final layer and implicitly assumes that the output probabilities faithfully reflect the reliability of the whole decision process. The paper challenges precisely this assumption and argues that such a view is too restrictive. Beyond this confidence baseline, the authors position their work relative to several topological approaches to neural network analysis.

1.3 Presentation of the proposed method

1.3.1 General idea

The main idea of the paper is to evaluate uncertainty by looking at how a new input activates the whole network, rather than only inspecting the final softmax layer. For a given sample, the network induces a collection of activation graphs, one for each layer transition. These graphs encode which neurons are effectively active and how strongly they contribute to the forward pass.

The method then compares the activation pattern of a new sample with the typical activation patterns observed on the training set. If the new sample activates the network in a way that is close to what was seen during training, then the prediction is considered more reliable. Conversely, if the activation pattern is atypical, the sample is assigned a larger uncertainty score, even if the final softmax confidence is high.

The topological ingredient of the paper is to summarize each activation graph through a persistence-based descriptor. This allows the authors to extract a compact but structured representation of how information flows through the network. The resulting uncertainty score is called *Topological Uncertainty* (TU) [2].

1.3.2 Mathematical Formulation: From Global Graphs to 1D Optimal Transport

To monitor a sequential neural network $F = f_L \circ \dots \circ f_1$, a natural geometric approach is to view the entire network as a single, complex topological space. Given an input x , one could construct a global activation graph representing all neurons and their weighted connections across all layers.

Persistent Homology and Optimal Transport. To extract meaningful features from such a graph, one relies on *Persistent Homology* (PH). PH studies the evolution of topological features such as connected components (0-dimensional homology) and loops (1-dimensional homology) across a sequence of nested subgraphs called a filtration. By progressively adding edges based on their activation weights, we record the "birth" and "death" of these topological features. These lifespans are summarized in a persistence diagram, a 2D point cloud where each point $(birth, death)$ represents a feature.

To compare how the network reacts to a new observation x versus a training sample x' , we must compute the distance between their respective persistence diagrams μ and ν . The standard mathematical tool for this is Optimal Transport (OT), specifically optimal partial matching metrics like the Wasserstein distance:

$$d_p(\mu, \nu) = \left(\inf_{\gamma} \sum_{x \in X \cup \Delta} \|x - \gamma(x)\|_1^p \right)^{1/p}$$

where γ ranges over all possible bijections between the points of the diagrams (including projections to the diagonal Δ).

The Computational Bottleneck. While theoretically rich, computing this OT distance on a large, global graph is a severe combinatorial problem. The algorithmic complexity of matching 2D point clouds makes it computationally prohibitive for real-time inference or monitoring in industrial applications.

Layer-wise Simplification and 1D Reduction. To bypass this bottleneck, Lacombe et al. introduce a radical but elegant simplification [2]. Instead of a single global graph, they restrict the analysis to bipartite graphs $G_\ell(x, F)$ connecting only two consecutive layers ℓ and $\ell + 1$. The weight of an edge between neuron i and neuron j is given by the activation strength:

$$w_\ell(i, j; x) = |W_\ell(i, j) x_\ell(i)|$$

Crucially, they restrict the topological analysis to 0-dimensional homology (connected components) using a superlevel set filtration. As edges are added in decreasing order of their weights, components merge, but no new vertices are ever created. In graph theory, tracking the merging of connected components in this manner is exactly equivalent to extracting a *Maximum Spanning Tree* (MST) of the graph.

Because every connected component eventually merges into a single one (death time at $+\infty$), the 2D coordinates $(w_i, +\infty)$ of the persistence diagram can be stripped of their second dimension. The diagram collapses into a purely 1-dimensional measure containing only the birth times (the MST edge weights):

$$\mu(G_\ell) = \frac{1}{N} \sum_{i=1}^N \delta_{w_i}$$

where $w_1 \geq w_2 \geq \dots \geq w_N$ and $N = d_\ell + d_{\ell+1} - 1$.

Trivial Optimal Transport in 1D. This 1D reduction is the key to computational efficiency. In Optimal Transport theory, computing the Wasserstein distance between two 1-dimensional empirical distributions of the same size is trivial: the optimal transport plan simply consists of matching the sorted points in increasing (or decreasing) order. Since the number of points N is fixed by the architecture of the layers, we completely avoid the complex 2D matching problem. The distance drastically simplifies to an L_1 norm between sorted vectors:

$$\text{Dist}(\mu, \nu) = \frac{1}{N} \sum_{i=1}^N |w_i - w'_i|$$

Topological Uncertainty. Finally, to assess if an input x is an Out-Of-Distribution (OOD) sample, we compare its fast-to-compute 1D diagrams to the average topological behavior of the training set. For each layer ℓ and each predicted class k , a Fréchet mean of the training diagrams, $D_{\ell,k}^{\text{train}}$, is stored. The Topological Uncertainty (TU) is then the average distance across all layers:

$$\text{TU}(x, F) = \frac{1}{L} \sum_{\ell=1}^L \text{Dist}\left(D_{\ell}(x, F), D_{\ell,k(x)}^{\text{train}}\right)$$

A high TU immediately flags an unusual activation pattern, circumventing the need for computationally heavy global graph comparisons.

1.4 Performance on Out-of-distribution detection

Although the paper evaluates Topological Uncertainty (TU) in several scenarios, we focus here only on out-of-distribution detection. The authors compare TU with the standard softmax-confidence baseline. They report result shown in Table 1. We see from them that TU consistently improves detection performance, with a particularly large gap on graph datasets. Even on image datasets, where the gain is less spectacular, TU still provides a significantly more informative signal than confidence alone.

Table 1: OOD detection performance of the softmax-confidence baseline and Topological Uncertainty (TU)

Training data	OOD data	Baseline AUC(%) $\uparrow$	TU AUC (%) $\uparrow$
MUTAG	Fake-MUTAG	1.7	99.8
COX2	Fake-COX2	1.4	99.9
CIFAR-10	Gaussian	27.2	88.2

These results support the main claim of the paper: atypical samples may still receive very high softmax confidence, whereas TU is much more sensitive to abnormal internal activation patterns. By using topological summaries of activation graphs throughout the network, Topological Uncertainty provides a stronger and more robust post-training indicator of reliability..

2 Our work

The public repository released by the authors mainly focuses on graph datasets and simple synthetic examples such as two moons. In our case, we decided to focus instead on image data, since this is the setting in which we have the most knowledge.

2.1 Reimplementation for image OOD detection

As a first step, we reimplemented a image OOD detection pipeline from scratch. We implemented both:

- the **standard baseline**, based only on the output layer and the prediction confidence,
- and the **Topological Uncertainty (TU)** score proposed in the paper.

The idea was the following: starting from a trained image classifier, we compute the baseline confidence score from the final output layer, and in parallel we compute the TU score by placing hooks on several internal layers of the network and extracting the corresponding topological summaries.

2.2 First experiments with a pretrained ResNet

In the paper, the authors state that they use a standard CNN trained with Adam for CIFAR-10. However, we chose to start with a **pretrained ResNet-56 on CIFAR-10**.

As discussed in the introduction, OOD detection becomes especially relevant when one has a classifier that performs well on in-distribution training data, but whose decision should not be trusted when the input is out of distribution is then used "in the real world". From this point of view, using a weak or relatively underperforming architecture is not fully convincing (Their standard CNN is really small and not that good). In many practical situations, the question is not whether a badly trained classifier fails on unusual inputs; rather, it is whether a strong classifier can still produce unreliable but confident predictions on OOD or shifted samples.

2.3 Baseline discrepancy

When reading the paper cited by the authors for the baseline, we noticed that the reported numbers were very different from those shown in the TU paper.

Indeed, in the TU paper, the baseline on CIFAR-10 versus Gaussian OOD is reported with an AUC-ROC around 27, which suggests extremely poor performance. However, when going back to the original baseline reference, the reported numbers were much higher, closer to 90 in comparable settings. This immediately raised concerns for us regarding the image experiments.

Because the source code for the image experiments is not provided, it is difficult to understand exactly how the baseline was implemented, and why its performance is so low in their reported table. This is problematic, because it makes the proposed method look much better than a very simple baseline, whereas in practice a simple baseline can already work reasonably well.

2.4 Experiments with a CNN closer to the paper

Since the paper explicitly states that the CIFAR-10 experiments were performed with a standard CNN, we also re-ran the experiments with a CNN architecture close to the one described in the paper.

This was important for two reasons:

1. to get closer to their stated experimental setting,
2. and to check whether part of the discrepancy could simply come from the architecture choice.

2.5 Layer-weight experiments

One interesting feature of TU is that it can be computed from several internal layers and then aggregated. This also allows us to test whether OOD information is concentrated more in low or high level layers.

For this reason, beyond the full TU score, we also performed experiments in which TU is computed only from:

- the early layers of ResNet-56,
- or only from the late layers of ResNet-56.

The intuition is the following. Early layers tend to capture more local and low-level features such as edges, textures, or simple color patterns, whereas later layers capture more semantic and task-specific features. By restricting TU to one part of the network, we can study whether topological deviations caused by OOD inputs are mostly visible at a low-level representation stage or at a higher-level semantic stage.

This also gives a first interpretability angle: if early-layer TU is already sufficient to detect OOD samples, then the abnormality may come from low-level statistics; if late-layer TU is more informative, then the mismatch may be more semantic and task-dependent.

2.6 Quantitative results

Model / Setting	Accuracy	Baseline AUC-ROC	TU AUC-ROC
CNN-like model	0.7141	0.8238	1
ResNet-56 (all selected layers)	0.9425	0.7690	0.5506
ResNet-56 (early layers only)	0.9425	0.7697	0.2639
ResNet-56 (late layers only)	0.9425	0.7714	0.7245

Table 2: OOD detection results on CIFAR-10.

Overall, our results are much closer to what is reported in the [1] than to what is reported for TU in the paper under study; see Section 3 for a more detailed discussion of this discrepancy. In particular, the confidence-based baseline already achieves reasonably strong OOD detection performance in our experiments, which contrasts with the very low baseline numbers reported in the paper.

Globally, the results show two very different behaviors depending on the architecture. For the **CNN-like model**, TU achieves an almost perfect separation between in-distribution and OOD samples, while the baseline is already strong but clearly lower. For the **ResNet-56**, however, the picture is very

different: the baseline remains around 0.77 AUC-ROC, while TU performs significantly worse overall, except when only late layers are used.

It is also worth noting that the three baseline AUC values reported for the ResNet are slightly different, even though the baseline itself should in principle not depend on the TU layer selection. This is simply due to the fact that the experiments were run on slightly different sampled subsets, so these small variations should not be over-interpreted. One advantage of the TU framework, however, is precisely that it allows us to probe where the useful information is located inside the network.

From this point of view, the early/late layer comparison is particularly informative. We observe that TU computed only on the early layers of the ResNet performs very poorly, which suggests that the first layers do not capture the relevant information needed to distinguish CIFAR-10 from Gaussian OOD samples. In contrast, restricting TU to the late layers gives much better results, although it still does not match the baseline. This indicates that, in this setting, the most useful OOD-related information is concentrated in the deeper and more semantic parts of the network rather than in the low-level feature extractors.

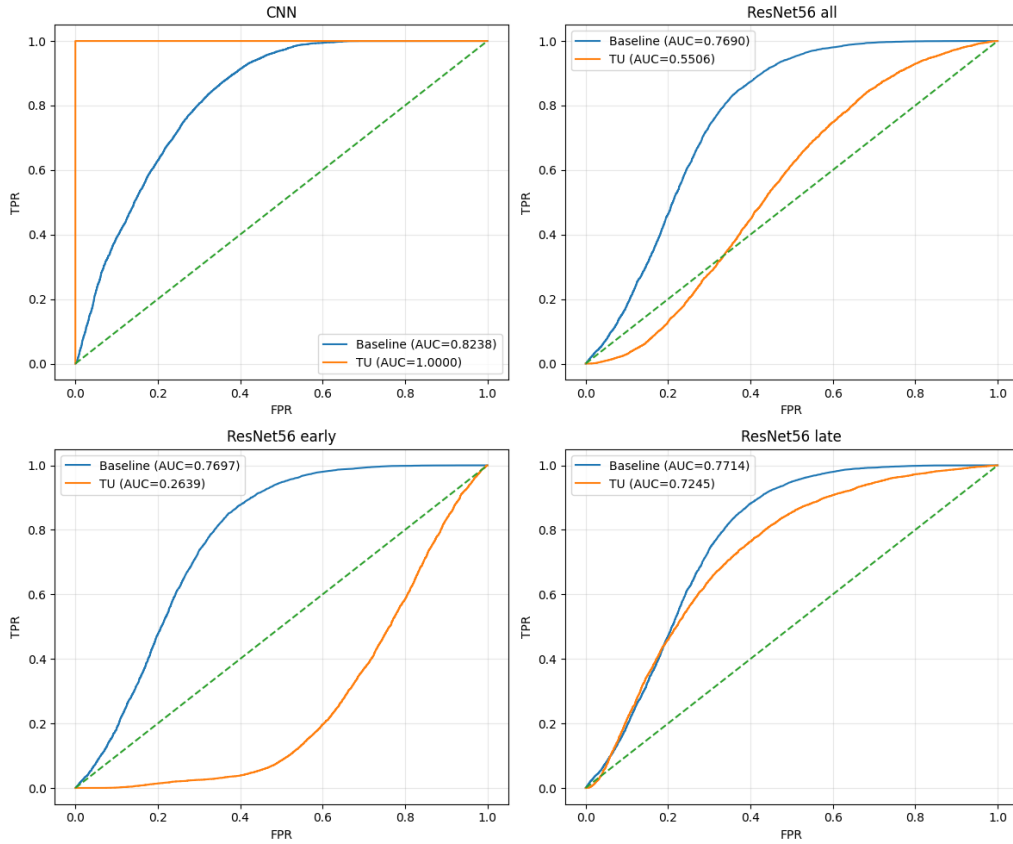


Figure 1: ROC curves for the baseline and TU scores on the different experimental settings.

2.7 Application on a Complex Use Case: Vehicle Detection via Drone Imagery

To qualitatively and quantitatively evaluate the Topological Uncertainty (TU) method on a complex, real-world problem, we conducted a series of experiments on an object detection task. The objective is to study the behavior of a trained neural network when confronted with domain shifts (out-of-distribution data) that were not encountered during its training phase.

2.7.1 Dataset and Model Presentation

We utilized a model from the YOLO family, which is widely recognized for its real-time object detection capabilities. This network was fine-tuned on the **VisDrone** reference dataset. Although YOLO jointly

solves a bounding box regression problem and a classification problem, our study strictly isolates its classification confidence, specifically focusing on the "car" class.

To test the model’s robustness and evaluate the TU metric, we exposed the network to new, real-world drone footage that exhibits significant distribution shifts compared to the VisDrone baseline:

- **VisDrone (Train / In-Distribution):** Features standard drone views at moderate altitudes with varied camera angles (often oblique), capturing standard vehicle profiles.
- **Medium Altitude (Moderate Shift):** Features nadir (strict top-down) views at a medium altitude. The perspective changes entirely, removing lateral features like doors or grilles.
- **High Altitude (Severe Shift):** Features nadir views at a high altitude. Vehicles are reduced to mere clusters of pixels, suffering from extreme loss of geometric and textural detail.



Figure 2: (a) Standard VisDrone with moderate altitude and varied angles, (b) Medium Altitude with a strict top-down (nadir) view, and (c) High Altitude where vehicles are reduced to small pixel clusters.

2.7.2 Selection of Analyzed Layers

Again, a core premise of the studied paper is that monitoring the entire network provides a better safety mechanism than looking only at the final output. Therefore, rather than extracting a single global graph, we isolated the activation graphs at key depths of the YOLO architecture (specifically Layers 2, 4, and 6) to track the evolution of the topological representations. In the context of YOLO’s architecture, these layers correspond to specific functional feature-extraction blocks:

- **Layer 2 (Low-level features / Early Backbone):** Typically corresponding to early Convolutional blocks in the backbone. It processes primary visual features such as raw edges, local contrasts, and basic textures directly extracted from the input image.
- **Layer 4 (Mid-level features / Mid-Backbone):** Corresponding to deeper convolutional feature extraction blocks. It aggregates these primary features into more complex structural motifs, such as windshield shapes, roof rectangles, or wheels.
- **Layer 6 (High-level Semantic features / Deep Backbone or Neck):** Associated with the final stages of the backbone, Spatial Pyramid Pooling block. It contains deep, highly abstract representations that are directly responsible for routing the final classification and bounding box regression decisions.

To visually quantify this phenomenon, Figure 3 illustrates the Topological Uncertainty (TU) density distributions for the three datasets across these selected layers.

2.7.3 Quantitative Analysis: Topological Uncertainty vs. Confidence

We first investigated the correlation between the network’s classical Confidence (the softmax probability) and the Topological Uncertainty (TU) across each selected layer.

Although TU scores are not normalized across layers, their relative evolution reveals a clear trend. In early blocks (Layers 2 and 4), TU scores for both *Medium* and *High* altitude datasets remain close to the training baseline. However, in the deeper semantic blocks (Layer 6), the *High Altitude* TU increases,

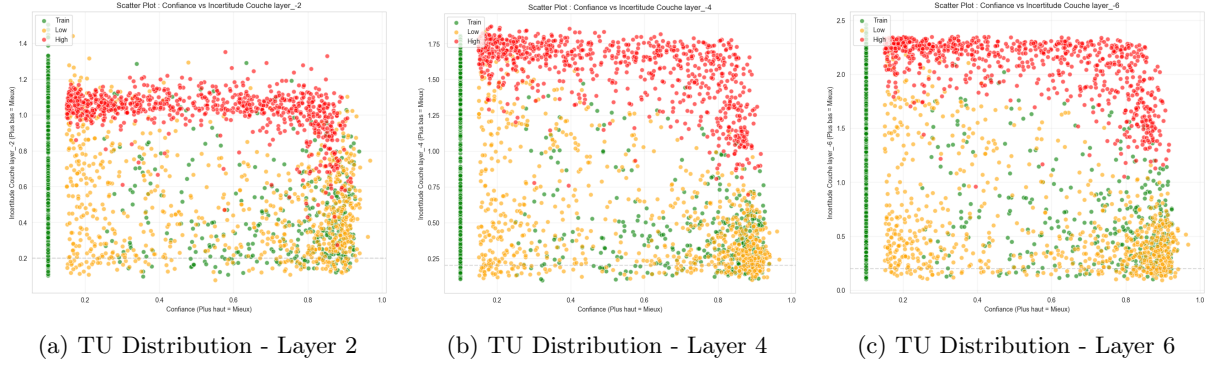


Figure 3: Evolution of Topological Uncertainty (TU) density distributions across the network’s depth.

decoupling from the other distributions. Conversely, the network’s standard confidence remains as high for these predictions. This perfectly illustrates that neural networks can be wildly overconfident on shifted-distribution data, and the final softmax layer alone is insufficient to flag severe domain shifts in this case, though TU is not perfect either.

2.7.4 Qualitative Evaluation: The Value of TU on Predictions

To ground these mathematical observations, we isolated specific predictions made by the network on the High Altitude dataset (Figure 4). We filtered for bounding boxes classified as ”car” with high confidence, and divided them into two distinct groups based on their TU scores:

- **High Confidence + High TU :** Visual inspection reveals that these qualitative results are poor. The model detects non-car objects anomalies with good certainty. Here, the high TU successfully acts as a crucial safety alarm: it indicates that the sequence of neuronal activations leading to this confident decision is topologically strange compared to how the network processes a real car.
- **High Confidence + Low TU:** When the TU remains low, visual inspection confirms that the network accurately target actual vehicles. The prediction is correct, and the activation path safely mirrors the learned training distribution. (Note: We empirically observe a slight bias here. The network tends to assign these ”perfect” low-TU/high-confidence scores almost exclusively to white vehicles).

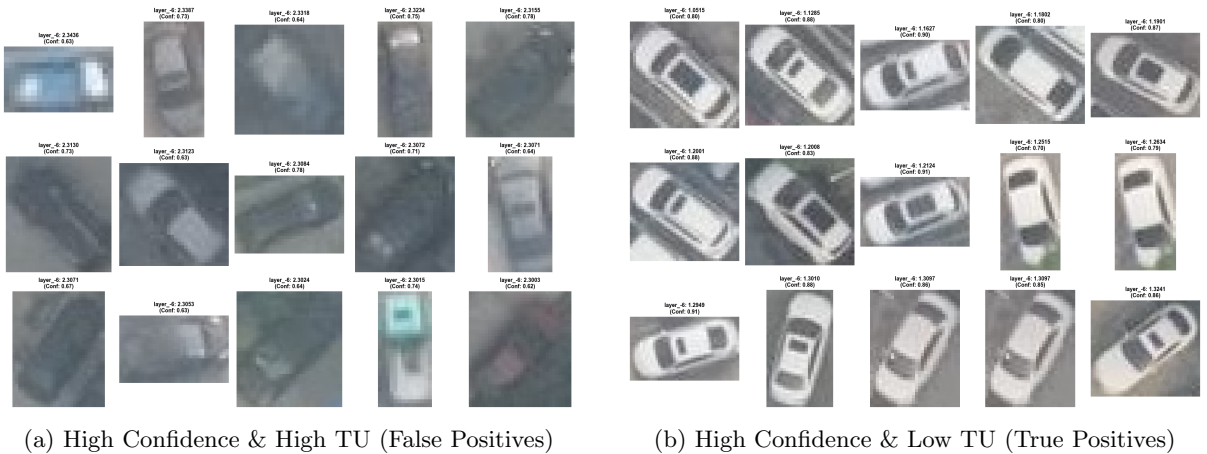


Figure 4: Left: Examples of image patches with High Confidence and High TU. Right: Patches with High Confidence and Low TU.

3 Critical Analysis and Extended Evaluation

We have so far mainly focused on reproducing the method and reporting our own experiments. However, we have somewhat brushed aside several important issues, so we conclude with a short critique of the paper, divided into methodological shortcomings, the method’s practical potential, and our final verdict.

3.1 Methodological and Experimental Shortcomings

First, the lack of source code for the image experiments is a serious issue. The public repository mainly covers graphs and simple toy datasets. Without code, it is difficult to verify whether the baseline was implemented fairly and whether the reported image results are really reproducible.

Second, the way the baseline is presented in the paper is, in our view, somewhat misleading. The reported baseline performance on CIFAR-10 versus Gaussian OOD is extremely low, which makes the proposed method appear dramatically superior. But when going back to the original baseline reference [1], the numbers seem much stronger. This suggests that the comparison may not be as clear-cut as the paper makes it look.

Third, the architecture used by the authors for CIFAR-10 is a relatively standard CNN. From our point of view, this is not the most convincing setting for OOD detection. In practice, one is mostly interested in monitoring models that already perform well and that are actually deployed. This is why we believe that testing the method on a stronger architecture such as ResNet-56 is more informative.

3.2 Practical Value and Extended Evaluation

That being said, our criticism is not that the TU score is uninteresting. On the contrary, we think the main idea is strong. Looking beyond the output confidence and studying the internal activation structure of the network is a meaningful direction. In particular, TU seems to provide information that is different from standard confidence, and this suggests that a combination of both could be more useful than either one alone.

To ground this intuition in a practical setting and further nuance our critique, we extended the evaluation to a highly complex, real-world use case: vehicle detection via drone imagery using a deployed YOLO model. In this operational scenario, we face a genuine industrial challenge where the annotated training data (standard VisDrone views) differs significantly from the data encountered during inference (medium and high-altitude nadir views).

Our qualitative analysis confirms that while the Topological Uncertainty is not a flawless metric that resolves all detection issues, it exhibits clear practical value. As the domain shift increases in severity, the TU score distinctly rises, capturing the internal deviation of the network’s activations. Therefore, TU serves as a highly relevant early-warning system. It effectively flags when deployed models are operating on shifted data, warning practitioners that the current activation pathways have strayed too far from the learned training distribution, even when the final softmax output remains blindly overconfident.

3.3 Future Directions and Open Questions

While our evaluation demonstrates the practical value of Topological Uncertainty (TU), this study also opens for future research to refine and scale the method:

- **Validity of the Fréchet Mean:** If the intra-class variance of activation diagrams is high, measuring the distance to a single mean $D_{\ell,k}^{\text{train}}$ might be overly reductive. Future work should quantify this variance and explore multi-modal topological representations (e.g., clustering diagrams within a single class). Moreover in our current code we have as a hyperparameter the number of sample per class to compute the Fréchet mean because it is quite computationally heavy, we could discuss this parameter.
- **Global Graph & Sliced OT:** Computing persistent homology over the entire activation graph rather than layer-by-layer could capture complex, cross-layer features. To avoid the big computational cost of general Optimal Transport, fast approximations like the Sliced Wasserstein Distance (Sliced OT) could bridge the gap between theoretical richness and computational feasibility.
- **Robust Normalization:** TU values currently depend heavily on the chosen architecture and layer dimensions. Developing a mathematically sound normalization scheme would transform TU into a universal, easily interpretable metric that can be directly compared across different models.

- **Alternative Modalities:** Extending the evaluation beyond images and simple graphs to test TU on sequential data (e.g., NLP tasks using Transformers) or 3D point clouds would further validate its robustness and generalizability as an Out-Shifted-Distribution / Of-Distribution detector.

References

- [1] Dan Hendrycks and Kevin Gimpel. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. In *International Conference on Learning Representations (ICLR)*, 2017.
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